t-STRUCTURES

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Contents

The purpose of these notes is to concisely define and discuss basic properties of t-structures and the perverse t-structure of complexes of constructible ℓ-adic sheaves

1. Triangulated categories

Definition 1.0.1 (Triangulated category)**.** A *triangulated category* is a triple $(\mathcal{T}, [1], \Delta)$ where:

- $\mathcal{T}$ is an additive category,
- $[1] : \mathcal{T} \to \mathcal{T}$ (also denoted by notations such as Σ or T) is an additive auto-equivalence called the *shift* or *suspension functor*,
- Δ is a class of *distinguished triangles* of the form

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

with objects $X, Y, Z \in \mathcal{T}$ and morphisms f, g, h in $\mathcal{T}$, satisfying the following axioms:

(TR1) The class Δ is closed under isomorphisms of triangles and contains the triangles isomorphic to

$$A \xrightarrow{\mathrm{id}} A \to 0 \to A[1].$$

Moreover, for every morphism $f : X \to Y$ in $\mathcal{T}$, there exists a distinguished triangle

$$X \xrightarrow{f} Y \to Z \to X[1].$$

(TR2) (Rotation) If

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

is a distinguished triangle, then so are

$$Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$$

and

$$Z[-1] \xrightarrow{-h[-1]} X \xrightarrow{f} Y \xrightarrow{g} Z.$$

(TR3) (Octahedral axiom) Given morphisms $X \xrightarrow{f} Y$ and $Y \xrightarrow{g} Z$ in $\mathcal{T}$, there exist distinguished triangles

$$X \xrightarrow{f} Y \xrightarrow{u} C(f) \xrightarrow{v} X[1],$$

$$Y \xrightarrow{g} Z \xrightarrow{w} C(g) \xrightarrow{x} Y[1],$$

$$X \xrightarrow{g\circ f} Z \xrightarrow{y} C(g \circ f) \xrightarrow{z} X[1],$$

and morphisms

$$s : C(f) \to C(g \circ f), \quad t : C(g \circ f) \to C(g)$$

such that the following diagram commutes and its rows and columns are distinguished triangles:

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, equal] & Y \arrow[r, "u"] \arrow[d, "g"] & C(f) \arrow[r, "v"] \arrow[d, dashed, "s"] & X[1] \arrow[d, equal] \\
X \arrow[r, "g\circ f"] & Z \arrow[r, "y"] \arrow[d, "w"] & C(g \circ f) \arrow[r, "z"] \arrow[d, dashed, "t"] & X[1] \\
 & C(g) \arrow[r, equal] \arrow[d, "x"] & C(g) \arrow[d] & \\
 & Y[1] \arrow[r, "f[1]"] & X[1][1] &
\end{tikzcd}$$

In particular,
- The compositions $s \circ u = y$, $t \circ s = w$, and $x \circ t = v[1]$ hold,
- All rows and columns form distinguished triangles.

(TR4) The class Δ is closed under the shift functor $[1]$: if

$$X \to Y \to Z \to X[1]$$

is distinguished, then so is

$$X[1] \to Y[1] \to Z[1] \to X[2].$$

Definition 1.0.2. Let $\mathcal{T}$ be a triangulated category (Definition 1.0.1). For any morphism $u : X \to Y$ in $\mathcal{T}$, a *cone of u* is an object C that fits into a distinguished triangle (Definition 1.0.1):

$$X \xrightarrow{u} Y \xrightarrow{v} C \xrightarrow{d} X[1]$$

In a general triangulated category, the cone C is guaranteed to exist by the axioms, but it is only unique up to a (not necessarily unique) isomorphism.

Definition 1.0.3. Let $\mathcal{T}$ be a triangulated category (Definition 1.0.1). Let (X, Y, Z, u, v, d) and (X', Y', Z', u', v', d') be two distinguished triangles (Definition 1.0.1) in $\mathcal{T}$. A *morphism of triangles* is a triple (f, g, h) of morphisms $f : X \to X'$, $g : Y \to Y'$, and $h : Z \to Z'$ such that the following diagram commutes:

$$\begin{tikzcd}
X \arrow[r, "u"] \arrow[d, "f"] & Y \arrow[r, "v"] \arrow[d, "g"] & Z \arrow[r, "d"] \arrow[d, "h"] & X[1] \arrow[d, "{f[1]}"] \\
X' \arrow[r, "u'"] & Y' \arrow[r, "v'"] & Z' \arrow[r, "d'"] & X'[1]
\end{tikzcd}$$

That is, the relations $g \circ u = u' \circ f$, $h \circ v = v' \circ g$, and $f[1] \circ d = d' \circ h$ must all hold.

Definition 1.0.4. A functor $T : D_1 \to D_2$ between triangulated categories (Definition 1.0.1) is called *exact* if it is additive, is translation preserving, and transforms distinguished triangles (Definition 1.0.1) to distinguished triangles. Oftentimes, a *morphism between triangulated categories* or a *triangulated functor* refers to an exact functor between triangulated categories.

Proposition 1.0.5 ([BBD82, Proposition 1.1.9.])**.** Let $\mathcal{T}$ be a triangulated category (Definition 1.0.1). Let (X, Y, Z, u, v, d) and (X', Y', Z', u', v', d') be two distinguished triangles (Definition 1.0.1) in $\mathcal{T}$. Let $g : Y \to Y'$ be a morphism in $\mathcal{T}$. Consider the following commutative diagram:

$$\begin{tikzcd}
X \arrow[r, "u"] \arrow[d, dashed, "f"] & Y \arrow[r, "v"] \arrow[d, "g"] & Z \arrow[r, "d"] \arrow[d, dashed, "h"] & X[1] \arrow[d, dashed, "{f[1]}"] \\
X' \arrow[r, "u'"] & Y' \arrow[r, "v'"] & Z' \arrow[r, "d'"] & X'[1]
\end{tikzcd}$$

The following conditions are equivalent:

(a) The composition $v' \circ g \circ u = 0$.
(b) There exists a morphism $f : X \to X'$ such that $g \circ u = u' \circ f$.
(b') There exists a morphism $h : Z \to Z'$ such that $h \circ v = v' \circ g$. (♠ TODO: morphism of distinguished triangles)
(c) There exists a morphism of triangles (f, g, h) from (X, Y, Z) to (X', Y', Z').

If these conditions are verified, and if $\operatorname{Hom}_{\mathcal{T}}(X, Z'[-1]) = 0$, the morphism f (resp. h) of (b) (resp. (b')) is unique.

Corollary 1.0.6 ([BBD82, Corollary 1.1.10.])**.** Let $\mathcal{T}$ be a triangulated category (Definition 1.0.1). Let $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{d} X[1]$ be a distinguished triangle (Definition 1.0.1) in $\mathcal{T}$. If $\operatorname{Hom}_{\mathcal{T}}(X, Z[-1]) = 0$, then:

(i) the cone of u (Definition 1.0.2) is unique up to unique isomorphism;
(ii) d is the unique morphism $x : Z \to X[1]$ such that the triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{x} X[1]$ is distinguished.

2. t-structures

See [BBD82, 1.3], [KW13, II.2-4, III.1].

D: A triangulated category.

T: The translation functor $T : D \to D$.

$A[1]$: $T(A)$ for $A \in D$.

Definition 2.0.1. A *strictly full subcategory of a category* is a subcategory that is full (Definition A.0.2) and also closed under isomorphism.

Definition 2.0.2 ([KW13, Definition II.2.1])**.** Let D be a triangulated category (Definition 1.0.1).

A t-structure on D consists of two strictly full subcategories (Definition 2.0.1) [1] $D^{\leq 0}$ and $D^{\geq 0}$ of D such that, writing

$$D^{\leq n} = D^{\leq 0}[-n]$$
$$D^{\geq n} = D^{\geq 0}[-n]$$

we have

1. $\operatorname{Hom}(D^{\leq 0}, D^{\geq 1}) = 0$,
2. $D^{\leq 0} \subset D^{\leq 1}$ and $D^{\geq 1} \subset D^{\geq 0}$, and
3. for every object E in D there exists a distinguished triangle (Definition 1.0.1) (A, E, B) with $A \in D^{\leq 0}$ and $B \in D^{\geq 1}$.

A *t-category* is a triangulated category equipped with a t-structure.

Given a t-structure t on D, the *core of t* is the full subcategory $\operatorname{Core}(D) = \operatorname{Core}_t(D) := D^{\leq 0} \cap D^{\geq 0}$ of D. It is an abelian category (Theorem 2.0.13).

(♠ TODO:) From now on, let $D^{\leq 0}, D^{\geq 0}$ be a t-structure on D unless otherwise specified.

[1]A *strictly full subcategory of a category* is a subcategory that is full and also closed under isomorphism.

Lemma 2.0.3. Let D be a triangulated category (Definition 2.0.2) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D. If $m, n \in \mathbb{Z}$ satisfy $m \leq n$, then

$$D^{\leq m} \supset D^{\leq n}$$
$$D^{\geq m} \subset D^{\geq n}$$

Proof. This is clear from the definitions and axioms on $D^{\leq n}, D^{\geq n}$. $\square$

Definition 2.0.4. Given a triangulated category (Definition 1.0.1) D with a t-structure (Definition 2.0.2) $(D^{\leq 0}, D^{\geq 0})$, call an object $M \in D$ *bounded* if $M \in D^{\geq a} \cap D^{\leq b}$ for some integers a and b. Say that D is *bounded with respect to the t-structure* if every object of D is bounded.

Proposition 2.0.5 ([BBD82, 1.3.3(i)])**.** Let D be a triangulated category (Definition 2.0.2) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D.

For all $n \in \mathbb{Z}$, the inclusion functor $D^{\leq n} \hookrightarrow D$ has a right adjoint (Definition A.0.1) $\tau_{\leq n} : D \to D^{\leq n}$ and the inclusion functor $D^{\geq n} \subset D$ has a left adjoint (Definition A.0.1) $\tau_{\geq n} : D \to D^{\geq n}$.

Lemma 2.0.6. Let D be a triangulated category (Definition 2.0.2) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D.

We have

$$\tau_{\leq n+1}(X) = (\tau_{\leq n}(X[1]))[-1]$$
$$\tau_{\geq n+1}(X) = (\tau_{\geq n}(X[1]))[-1].$$

Equivalently,

$$\tau_{\leq n}(X) = (\tau_{\leq 0}(X[n]))[-n]$$
$$\tau_{\geq n}(X) = (\tau_{\geq 0}(X[n]))[-n].$$

for all integer n.

Proof. Note that for $n = 0$, these are trivially true. Inductively suppose that these are true for $n = m$. We then have

$$\tau_{\leq m+1}(X) = \tau_{\leq m}(X[1])[-1] = (\tau_{\leq 0}(X[n+1])[-n])[-1] = \tau_{\leq 0}(X[n+1])[-n-1]$$
$$\tau_{\geq m+1}(X) = \tau_{\geq m}(X[1])[-1] = (\tau_{\geq 0}(X[n+1])[-n])[-1] = \tau_{\geq 0}(X[n+1])[-n-1]$$

and hence the desired equalities holds for all $n \geq 0$ by induction. Similar steps demonstrate the desired equalities for $n \leq 0$ as well. $\square$

From here on out, given a t-structure, fix such functors $\tau_{\leq 0}$ and $\tau_{\geq 1}$.

Proposition 2.0.7 ([BBD82, Proposition 1.3.3(ii)])**.** Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D.

For any object E in D, there exists a unique morphism $d \in \mathrm{Hom}(\tau_{\geq 1}E, \tau_{\leq 0}E[1])$ such that the triangle

$$\tau_{\leq E} \to E \to \tau_{\geq 1}E \xrightarrow{d} \tau_{\leq 0}E[1]$$

is distinguished (Definition 1.0.1). Up to unique isomorphism, every distinguished triangle (A, E, B) in D with $A \in D^{\leq 0}$ and $B \in D^{\geq 1}$ is equivalent to such a triangle.

Lemma 2.0.8. Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D.

We have

$$\begin{aligned} \tau_{\leq m} \circ \tau_{\leq n} = \tau_{\leq m} \quad &\text{for} \quad m \leq n \\ \tau_{\geq m} \circ \tau_{\geq n} = \tau_{\leq n} \quad &\text{for} \quad m \leq n. \end{aligned}$$

Proof. We prove the first statement. By Yoneda's lemma, it suffices to show that there are natural isomorphisms

$$\mathrm{Hom}_{D^{\leq m}}(A, \tau_{\leq m}\tau_{\leq n}B) \cong \mathrm{Hom}_{D^{\leq m}}(A, \tau_{\leq m}B)$$

for all objects $A \in D^{\leq m}$ and $B \in D$. By Proposition 2.0.5, the left hand side is isomorphic to $\mathrm{Hom}_D(A, \tau_{\leq n}B)$. As $A \in D^{\leq m} \subset D^{\leq n}$, $B \in D^{\leq n}$, and $D^{\leq n}$ is a full subcategory of D, the Hom-group $\mathrm{Hom}_D(A, \tau_{\leq n}B)$ is identifiable with the Hom-group $\mathrm{Hom}_{D^{\leq n}}(A, \tau_{\leq n}B)$. In turn, this Hom-group is identifiable with $\mathrm{Hom}_D(A, B)$, which is identifiable with $\mathrm{Hom}_{D^{\leq m}}(A, \tau_{\leq m}B)$ as desired by the adjunction of $\tau_{\leq n}$.

□

Lemma 2.0.9 (See [KW13, Corollary II.1.6])**.** Let D be a triangulated category (Definition 1.0.1). Let (X, Y, Z, u, v, w) be a distinguished triangle (Definition 1.0.1) of D. The morphism u is an isomorphism if and only if $Z \cong 0$.

Lemma 2.0.10 (Orthogonality, see [KW13, Lemma II.2.2])**.** Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D. Let $E \in D$. The following are equivalent:

1. $E \in D^{\geq n+1}$
2. $\mathrm{Hom}(D^{\leq n}, E) = 0$.

The following are equivalent:

1. $E \in D^{\leq n-1}$
2. $\mathrm{Hom}(E, D^{\geq n}) = 0$.

Proof. We show the former equivalence for $n = 0$ — the dual equivalence and both equivalences for general n can be easily deduced. Note that (2) immediately implies (1) by the axioms of a t-structure. Conversely, we first deduce that $\tau_{\leq 0}(E) = 0$ by the adjunction isomorphism and Lemma 2.0.8:

$$\mathrm{Hom}_{D^{\leq 0}}(\tau_{\leq 0}(E), \tau_{\leq 0}(E)) \cong \mathrm{Hom}_D(\tau_{\leq 0}\tau_{\leq 0}(E), E) \subset \mathrm{Hom}_D(\tau_{\leq 0}(E), E) = 0.$$

Taking the distinguished triangle

$$\tau_{\leq 0}(E) \to E \to \tau_{\geq 1}(E) \to \tau_{\leq 0}(E)[1]$$

of Proposition 2.0.7 and rotating it appropriately, Claim 2.0.9 shows that $E \cong \tau_{\geq 1}(E)$ and hence $E \in D^{\geq n+1}$. □

Lemma 2.0.11 (Extensions, see [KW13, Lemma II.2.3])**.** Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D.

Let (X, Y, Z) be a distinguished triangle. then

1. If $X, Z \in D^{\leq n}$, then $Y \in D^{\leq n}$ as well.
2. If $Y \in D^{\leq n}$ and if $X \in D^{\leq n+1}$, then $Z \in D^{\leq n}$.

Proof. (♠ TODO: TODO: prove) The long exact sequence

$$\cdots \to \mathrm{Hom}(X[n+1], A) \to \mathrm{Hom}(Z[n], A) \to \mathrm{Hom}(Y[n], A) \to \mathrm{Hom}(X[n], A) \to \cdots$$

$$\cdots \to \mathrm{Hom}(A, X[n]) \to \mathrm{Hom}(A, Y[n]) \to \mathrm{Hom}(A, Z[n]) \to \mathrm{Hom}(A, X[n+1]) \to \cdots$$

□

Lemma 2.0.12 (Compatibility, see [KW13, Lemma II.2.4])**.** Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D.

We have

$$\tau_{\geq m}(D^{\leq n}) \subset D^{\leq n}$$
$$\tau_{\leq m}(D^{\geq n}) \subset D^{\geq n}$$

(Proposition 2.0.5) for all $m, n \in \mathbb{Z}$. We have

$$\tau_{\geq m}(D^{\leq n}) = \tau_{\leq n}(D^{\geq m}) = 0$$

for $m > n$.

Proof. We prove the first inclusion. For $n < m$ and $X \in D^{\leq n}$ the unit map $X \to \tau_{\geq m}X$ of the adjunction for $\tau_{\geq m}$ is zero as a consequence of the axioms of a t-structure. The adjunction then implies that $\mathrm{Hom}(\tau_{\geq m}X, \tau_{\geq m}X) = 0$, so $\tau_{\geq m}X = 0$.

Now suppose that $m \leq n$. If $X \in D^{\leq n}$, then $\tau_{\leq m-1}(X) \in D^{\leq m-1} \subset D^{\leq n+1}$. In the distinguished triangle $(\tau_{\leq m-1}X, X, \tau_{\geq m}X)$ and applying the extensions Lemma 2.0.11, we have that $\tau_{\geq m}X \in D^{\leq n}$.

Similarly, the second incluskion holds. Note that we already proved the vanishing. □

Theorem 2.0.13 (See [KW13, Theorem II.3.1])**.** Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D. The core $\mathrm{Core}(D)$ (Definition 2.0.2) is an abelian category. A sequence

$$0 \to X \xrightarrow{u} Y \xrightarrow{v} Z \to 0$$

is exact if and only if there exists a distinguished triangle (Definition 1.0.1) (X, Y, Z, u, v, w) in D.

3. Cohomology functors for a t-structure

Definition 3.0.1. Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D. Define the *cohomology functors* $H^n : D \to \mathrm{Core}(D)$ as follows:

$$H^0(X) = \tau_{\leq 0}\tau_{\geq 0}X$$
$$H^n(X) = H^0(X[n]).$$

Note that $H^0(X) \in \mathrm{Core}(D)$ by Lemma 2.0.12.

Lemma 3.0.2. Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D. Let $X \in D$.

We have

1. $H^a(X[b]) = H^{a+b}(X) = H^0(X[a+b])$ (Definition 3.0.1) for all $a, b \in \mathbb{Z}$, and
2. $H^n(X) = (\tau_{\leq n}\tau_{\geq n}X)[n]$ (Proposition 2.0.5) (Definition 1.0.1) for all $n \in \mathbb{Z}$.

Proof. 1. Observe that

$$H^a(X[b]) = H^0((X[b])[a]) = H^0(X[a+b]) = H^{a+b}(X).$$

2. The left hand side equals

$$H^0(X[n]) = \tau_{\leq 0}\tau_{\geq 0}(X[n]),$$

whereas the right hand side equals

$$\begin{aligned}\tau_{\leq n}((\tau_{\geq 0}(X[n]))[-n])[n] &= \tau_{\leq 0}(\tau_{\geq 0}(X[n]))[-n][n] \\ &= \tau_{\leq 0}(\tau_{\geq 0}(X[n]))\end{aligned}$$

by Lemma 2.0.6.

□

Theorem 3.0.3 (See [KW13, Theorem 4.4])**.** Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D. Let (X, Y, Z) be a distinguished triangle in D. The cohomological functors (Definition 3.0.1) attached to a t-structure of D induce a long exact cohomology sequence

$$\cdots \to H^{-1}(Z) \to H^0(X) \to H^0(Y) \to H^0(Z) \to H^1(X) \to \cdots$$

in $\mathrm{Core}(D)$ (Definition 2.0.2).

Definition 3.0.4. Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D. The t-structure $(D^{\leq 0}, D^{\geq 0})$ is said to be *non-degenerate* if $\bigcap_n D^{\leq n}$ and $\bigcap_n D^{\geq n}$ each only have zero objects.

Proposition 3.0.5 ([BBD82, Proposition 1.3.7])**.** Let D be a triangulated category (Definition 1.0.1) and let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D.

(♠ TODO: conservative system of functors) If the t-structure $(D^{\leq 0}, D^{\geq 0})$ on D is non-degenerate (Definition 3.0.4), then the system of functors H^n is conservative, i.e. for a morphism $f : M \to N$ between objects of D, the induced maps $H^n(f) : H^n(M) \to H^n(N)$ are all isomorphisms if and only if f is an isomorphism. Moreover, an object X in D belongs to $D^{\leq 0}$ (resp. $D^{\geq 0}$) if and only if the $H^n(X)$ are zero for $n > 0$ (resp. for $n < 0$).

4. Recollement of t-structure

The following from [BBD82, Section 1.4] describes a general framework for gluing t-structures on triangulated categories.

Context 4.0.1 ([BBD82, 1.4.3.])**.** Let D, D_U, and D_F be triangulated categories (Definition 1.0.1). Suppose that there exist exact functors (Definition 8.0.1)

$$D_F \xrightarrow{i_*} D \xrightarrow{j^*} D_U.$$

[BBD82] sets $i_! := i_*$ and $j^! := j^*$ for convenience. Assume that the following hold:

1. i_* has exact left and right adjoints, denoted by i^* and $i^!$ respectively. In other words, we have adjunctions
$$i^* \dashv i_* \dashv i^!$$
2. j^* has exact left and right adjoints, denoted by $j_!$ and j_* respectively. In other words, we have adjunctions
$$j_! \dashv j^* \dashv j_*$$
3. $j^* i_* = 0$. By adjunction, we consequently have $i^* j_! = 0$, $i^! j_* = 0$, and for all $A \in D_F$ and $B \in D_U$,
$$\mathrm{Hom}_D(j_! B, i_* A) = 0 \quad \text{and} \quad \mathrm{Hom}_D(i_* A, j_* B) = 0.$$
4. For all K in D there exists $d : i_* i^* K \to j_! j^* K[1]$ (resp. $d : j_* j^* K \to i_* i^! K[1]$), which necessarily must be unique by the above condition and Corollary 1.0.6, such that the triangle
$$j_! j^* K \to K \to i_* i^* K \xrightarrow{d}$$
$$(\textit{resp.} \quad i_* i^! K \to K \to j_* j^* K \xrightarrow{d})$$
is distinguished.
5. i_*, $j_!$, and j^* are fully faithful (Definition A.0.2), i.e. the adjunction morphisms $i^* i_* \to \mathrm{id} \to i^! i_*$, and $j_! j^* \to \mathrm{id} \to j_* j^*$ are isomorphisms.

This formalism is self-dual — the duality exchanges $j_!$ and j_* and i^* and $i^!$.

Definition 4.0.2 ([BBD82, 1.4.9.])**.** Assume Context 4.0.1. Let $(D_U^{\leq 0}, D_U^{\geq 0})$ and $(D_F^{\leq 0}, D_F^{\geq 0})$ be t-structures (Definition 2.0.2) on D_U and D_F respectively. Define

$$D^{\leq 0} := \{K \in D \mid j^*K \in D_U^{\leq 0} \text{ and } i^*K \in D_F^{\leq 0}\}$$
$$D^{\geq 0} := \{K \in D \mid j^*K \in D_U^{\geq 0} \text{ and } i^!K \in D_F^{\geq 0}\}$$

It is a t-structure on D (Theorem 4.0.3). We might call this t-structure the one *induced from those of D_U and D_F by gluing or recollement*.

Theorem 4.0.3 ([BBD82, Théorème 1.4.10., Remarque 1.4.11.])**.** Assume Context 4.0.1. Let $(D_U^{\leq 0}, D_U^{\geq 0})$ and $(D_F^{\leq 0}, D_F^{\geq 0})$ be t-structures (Definition 2.0.2) on D_U and D_F respectively. The pair $(D^{\leq 0}, D^{\geq 0})$ is a t-structure (Definition 2.0.2) on D. Moreover, $(D^{\leq 0}, D^{\geq 0})$ is non-degenerate (Definition 3.0.4) if and only if $(D_U^{\leq 0}, D_U^{\geq 0})$ and $(D_F^{\leq 0}, D_F^{\geq 0})$ both are.

Proposition 4.0.4 ([BBD82, Proposition 1.4.12.])**.** Assume Context 4.0.1. Let $(D^{\leq 0}, D^{\geq 0})$ be a t-structure (Definition 2.0.2) on D. The following are equivalent:

1. $j_!j^* : D \to D$ is right t-exact (Definition 8.0.2).
2. $j_*j^* : D \to D$ is left t-exact (Definition 8.0.2).
3. The t-structure of D is obtained by gluing t-structures on D_U and D_F.

More explicitly, if the above hold, then the t-structures on D_U and D_F are respectively $(D_U^{\leq 0}, D_U^{\geq 0})$ and $(D_F^{\leq 0}, D_F^{\geq 0})$ uniquely given by

$$D_U^{\leq 0} = j^*(D^{\leq 0})$$
$$D_U^{\geq 0} = j^*(D^{\geq 0})$$
$$D_F^{\leq 0} = i^*(D^{\leq 0})$$
$$D_F^{\geq 0} = i^!(D^{\geq 0})$$

5. Bounded derived category of constructible complexes of ℓ-adic sheaves

(♠ TODO:) From here, let X be a scheme over a field k of characteristic $p \geq 0$ and let $\ell \neq p$ be a prime number. There is a derived category $D_c^b(X) = D_c^b(X, \overline{\mathbb{Q}}_\ell)$ of bounded complexes of constructible sheaves with $\overline{\mathbb{Q}}_\ell$-coefficients. There is an associated six-functor formalism:

1. There are bifunctors $D_c^b(X) \to D_c^b(X) \to D_c^b(X)$ written
$$(M, N) \mapsto \mathrm{RHom}(M, N), \quad (M, N) \mapsto M \otimes N.$$
2. Given a k-morphism $f : X \to Y$, there are functors
$$f^* : D_c^b(Y) \to D_c^b(X)$$
and
$$Rf_* : D_c^b(X) \to D_c^b(Y).$$

3. If f is k-compactifiable (e.g. by virtue of Y being quasi-compact and quasi-separated and f being separated and of finite type), then there are functors $Rf_! : D_c^b(X) \to D_c^b(Y)$ and $f^! : D_c^b(Y) \to D_c^b(X)$.

These satisfy the following adjunctions (the last adjunction requires f to be k-compactifiable for $Rf_!$ to be defined):

$$\begin{aligned} \operatorname{Hom}(M \otimes N, P) &\cong \operatorname{Hom}(R, \operatorname{RHom}(N, P)) \\ \operatorname{Hom}(f^* M, N) &\cong \operatorname{Hom}(M, Rf_* N) \\ \operatorname{Hom}(Rf_! M, N) &\cong \operatorname{Hom}(M, f^! N). \end{aligned}$$

For any object $K \in D_c^b(X)$, there are the cohomology functors $K \mapsto \mathscr{H}^i(K)$. Note that each $\mathscr{H}^i(K)$ is a constructible sheaf with $\overline{\mathbb{Q}}_\ell$-coefficients.

6. Standard t-structure

Definition 6.0.1 (See [KW13, Definition II.6.3, Theorem II.6.4, and the discussion afterwards])**.** (♠ TODO:) The *standard t-structure on $D_c^b(X, \overline{\mathbb{Q}}_\ell)$* is

$$\begin{aligned} D^{\leq 0}(X, \overline{\mathbb{Q}}_\ell) &= \{K \in D_c^b(X) : \mathscr{H}^i(K) = 0 \text{ for all } i > 0\} \\ D^{\geq 0}(X, \overline{\mathbb{Q}}_\ell) &= \{K \in D_c^b(X) : \mathscr{H}^i(K) = 0 \text{ for all } i < 0\}. \end{aligned}$$

Write ${}^{\mathrm{st}}\tau_{\leq n}$ and ${}^{\mathrm{st}}\tau_{\geq n}$ for the truncation functors for this t-structure.

Claim 6.0.2. (♠ TODO:) There is a natural isomorphism

$$H^n(K) = {}^{\mathrm{st}}\tau_{\leq n} {}^{\mathrm{st}}\tau_{\geq n} K \cong \mathscr{H}^n(K)$$

for $K \in D_c^b(X)$ where $\mathscr{H}^n(K)$ here denotes the nth cohomology object with respect to the standard t-structure on $D_c^b(X)$.

7. Perverse t-structure

Now let X be a finite type and separated scheme over k.

Definition 7.0.1. (♠ TODO:) Let K_X denote the *dualizing complex of the X*, defined by

$$K_X = \pi^!(\overline{\mathbb{Q}}_\ell) \in D_c^b(X)$$

where $\pi : X \to \operatorname{Spec} k$ is the structure morphism and $\overline{\mathbb{Q}}_\ell$ here refers to the skyscraper sheaf with $\overline{\mathbb{Q}}_\ell$-coefficients of rank 1 on $\operatorname{Spec} k$. One defines the *Verdier dual* $D = D_X$ by

$$D_X(K) = \operatorname{RHom}(K, K_X).$$

Definition 7.0.2 (See [KW13, Section III.1]). (♠ TODO:) Let $D = D(X)$ be the triangulated category $D^b_c(X, \overline{\mathbb{Q}}_\ell)$. The *selfdual (middle) perverse t-structure* is defined by

$$B \in {}^pD^{\leq 0}(X) \Leftrightarrow \dim \operatorname{Supp}(\mathscr{H}^{-i}B) \leq i \quad \text{for all } i \in \mathbb{Z}$$
$$B \in {}^pD^{\geq 0}(X) \Leftrightarrow \dim \operatorname{Supp}(\mathscr{H}^{-i}DB) \leq i \quad \text{for all } i \in \mathbb{Z}$$

We say that $B \in D^b_c(X)$ is *semiperverse* if $B \in {}^pD^{\leq 0}(X)$.

Theorem 7.0.3 (See [KW13, Section III.3]). The perverse t-structure is indeed a t-structure.

Write $\operatorname{Perv}(X)$ for the core of the perverse t-structure and write ${}^p\mathscr{H}^i : D^b_c(X) \to \operatorname{Perv}(X)$ for the ith cohomology object with respect to this t-structure. Moreover, since the perverse t-structure is a genuine t-structure, we may write ${}^pD^{\leq n}, {}^pD^{\geq n}, {}^p\tau_{\geq n}, {}^p\tau_{\leq n}$, etc. for the relevant subcategories of $D^b_c(X)$ and the truncation functors.

Lemma 7.0.4. (♠ TODO:) For $B \in D^b_c(X)$,

$$B \in {}^pD^{\leq n}(X) \Leftrightarrow \dim \operatorname{Supp}(\mathscr{H}^{-i}B) \leq i + n \quad \text{for all } i \in \mathbb{Z}$$
$$B \in {}^pD^{\geq n}(X) \Leftrightarrow \dim \operatorname{Supp}(\mathscr{H}^{-i}DB) \leq i - n \quad \text{for all } i \in \mathbb{Z}$$

Proof. By definition, ${}^pD^{\leq n}(X) = {}^pD^{\leq 0}(X)[-n]$, so $B \in {}^pD^{\leq n}(X)$ if and only if $B[n] \in {}^pD^{\leq 0}(X)$. In turn, this is equivalent to

$$\dim \operatorname{Supp}(\mathscr{H}^{-i}(B[n])) \leq i \quad \text{for all } i \in \mathbb{Z}$$

or to

$$\dim \operatorname{Supp}(\mathscr{H}^{-i+n}B) \leq i \quad \text{for all } i \in \mathbb{Z}$$

by Lemma 3.0.2. Changing indices, this is equivalent to

$$\dim \operatorname{Supp}(\mathscr{H}^{-i}B) \leq i + n \quad \text{for all } i \in \mathbb{Z}$$

as desired. The second equivalence can be demonstrated similarly, taking into account that $D(B[n]) \cong (DB)[-n]$. □

Lemma 7.0.5. (♠ TODO:) For nonzero $B \in D^b_c(X)$,

1. $B \notin {}^pD^{\geq n}(X)$ for all $n \ll 0$.
2. $B \notin {}^pD^{\leq n}(X)$ for all $n \gg 0$.

Proof. The condition $B \in {}^pD^{\geq n}(X)$ is equivalent to the statement

$$\dim \operatorname{Supp}(\mathscr{H}^{-i}B) \leq i + n \quad \text{for all } i \in \mathbb{Z}$$

by Lemma 7.0.4. Since B is assumed to be nonzero, at least one of the $\mathscr{H}^{-i}B$ must be nonzero; the inequality $\dim \operatorname{Supp}(\mathscr{H}^{-i}B) \leq i + n$ cannot hold whenever $n < -i$. Therefore, B cannot be an object of ${}^pD^{\geq n}(X)$ for any such n. Similarly, $B \notin {}^pD^{\leq n}(X)$ for all $n \gg 0$ as well. □

Corollary 7.0.6. (♠ TODO:) The perverse t-structure is non-degenerate. In particular, for $B \in D^b_c(X)$, ${}^p\mathscr{H}^i(B) = 0$ for all $i \in \mathbb{Z}$ if and only if $B = 0$ and

$$ {}^pD^{\leq n}(X) = \{B \in D^b_c(X) : {}^p\mathscr{H}^i(B) = 0 \text{ for all } i > n\} $$
$$ {}^pD^{\geq n}(X) = \{B \in D^b_c(X) : {}^p\mathscr{H}^i(B) = 0 \text{ for all } i < n\}. $$

Proof. This is immediate from Lemma 7.0.5, the definition of a non-degenerate t-structure, and Proposition 3.0.5. □

Theorem 7.0.7 (See [KW13, Corollaries III.5.4, III.5.7])**.** (♠ TODO:) $\mathrm{Perv}(X)$ is artinian and noetherian. Moreover, $B \in \mathrm{Perv}(X)$ is simple if and only if it is of the form $B = j_{!*}\mathscr{G}[d]$ for a locally closed embedding $j : U \to X$ from an irreducible smooth d-dimensional subscheme U of dimension d and a lisse $\overline{\mathbb{Q}}_\ell$-sheaf $\mathscr{G}$.

Proposition 7.0.8 (See [KW13, Lemma III.5.13])**.** If $\dim(X) \leq d$, then ${}^pD^{\geq 0}(X, \overline{\mathbb{Q}}_\ell)$ is contained in ${}^{\mathrm{st}}D^{\geq -d}(X, \overline{\mathbb{Q}}_\ell)$.

8. Exactness properties for the perverse t-structure under direct images and inverse images

We discuss some conditions under which the derived functors $Rf_*, Rf_!, f^*, f^!$ (or variants thereof) preserve perversity.

Definition 8.0.1. A functor $T : D_1 \to D_2$ between triangulated categories (Definition 1.0.1) is called *exact* if it is additive, is translation preserving, and transforms distinguished triangles (Definition 1.0.1) to distinguished triangles. Oftentimes, a *morphism between triangulated categories* or a *triangulated functor* refers to an exact functor between triangulated categories.

Definition 8.0.2. Let $T : D_1 \to D_2$ be an exact functor (Definition 8.0.1) between triangulated categories (Definition 1.0.1) with t-structure (Definition 2.0.2). Such a functor is called *t-left exact* if $T(D_1^{\leq 0}) \subset D_2^{\leq 0}$, *t-right exact* if $T(D_1^{\geq 0}) \subset D_2^{\geq 0}$, and *t-exact* if T is both t-left and t-right exact.

Lemma 8.0.3 (cf. [Kat96, Section 2.1], [Kat90, Section 8.1],)**.** Let $f : X \to Y$ be a morphism of finite type schemes over a field k of characteristic distinct from ℓ.

1. ([BBD82, Théorème 4.1.1], cf. [DA73, Exposé XIV, Théorème], [KW13, Theorem III.6.1]) If $f : X \to Y$ is affine, then $Rf_* : D^b_c(X, \overline{\mathbb{Q}}_\ell) \to D^b_c(Y, \overline{\mathbb{Q}}_\ell)$ is right t-exact with respect to the perverse t-structures on $D^b_c(X, \overline{\mathbb{Q}}_\ell)$ and $D^b_c(Y, \overline{\mathbb{Q}}_\ell)$, i.e. restricts to a functor ${}^pD^{\leq 0}(X) \to {}^pD^{\leq 0}(Y)$, and $Rf_! : D^b_c(X, \overline{\mathbb{Q}}_\ell) \to D^b_c(Y, \overline{\mathbb{Q}}_\ell)$ is left t-exact. In particular, Rf_* preserves semiperversity.
2. (See [BBD82, Proposition 2.2.5]) If $f : X \to Y$ is a quasi-finite, then $Rf_!$ and f^* are right t-exact, and $f^!$ and Rf_* are left t-exact. In particular, $Rf_!$ and f^* preserve semiperversity.

Proposition 8.0.4. Let $f : X \to Y$ be a morphism of finite type schemes over a field k of characteristic distinct from ℓ.

1. If f is quasi-finite and affine, then $Rf_*M, Rf_!M \in \mathrm{Perv}(Y)$ for all $M \in \mathrm{Perv}(X)$ [KW13, Corollary III.6.2].
2. If f is equidimensional of relative dimension d and smooth, then
$$f^*[d](M) = f^![-d](-d)(M) \in \mathrm{Perv}(X)$$
for all $M \in \mathrm{Perv}(Y)$ [KW13, Theorem III.7.2].

Appendix A. Miscellaneous definitions

Definition A.0.1. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors.

An *adjunction between F and G* consists of two natural transformations: $\eta : \mathrm{Id}_{\mathcal{C}} \implies GF$ (the *unit*), and $\varepsilon : FG \implies \mathrm{Id}_{\mathcal{D}}$ (the *counit*)

These must satisfy the triangle identities: For every object $X \in \mathcal{C}$ and $Y \in \mathcal{D}$,
$$\varepsilon_{FX} \circ F(\eta_X) = \mathrm{id}_{FX}$$
$$G(\varepsilon_Y) \circ \eta_{GY} = \mathrm{id}_{GY}.$$
In diagrammatic form, the triangle identities assert that the following are commutative diagrams:

$$\begin{tikzcd} F(X) \arrow[r, "F(\eta_X)"] \arrow[dr, "\mathrm{id}_{F(X)}"'] & FGF(X) \arrow[d, "\varepsilon_{F(X)}"] & G(Y) \arrow[r, "\eta_{G(Y)}"] \arrow[dr, "\mathrm{id}_{G(Y)}"'] & GFG(Y) \arrow[d, "G(\varepsilon_Y)"] \\ & F(X) & & G(Y) \end{tikzcd}$$

We say that F is a *left adjoint to G* and G is a *right adjoint to F* (written $F \dashv G$).

In the case that $\mathcal{C}$ and $\mathcal{D}$ are locally small categories (or U-locally small categories if a universe U is available), we have an adjunction $F \dashv G$ if and only if for every object X in $\mathcal{C}$ and Y in $\mathcal{D}$ there is a natural isomorphism
$$\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$$
that is natural in both X and Y. More precisely, there are two functors
$$\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \to \mathbf{Sets}$$
given by
$$(X, Y) \mapsto \mathrm{Hom}_{\mathcal{D}}(F(X), Y)$$
$$(X, Y) \mapsto \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$$
respectively, and there is a natural isomorphism between them.

In this case, the *unit of the adjunction* is the natural transformation $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow GF$ such that,

1. for every $X \in \mathcal{C}$, the morphism $\eta_X : X \to GF(X)$ (each called a *unit morphism*) in $\mathcal{C}$ is obtained as the image of $\mathrm{id}_{F(X)}$ via the adjoint isomorphism
$$\mathrm{Hom}_{\mathcal{D}}(F(X), F(X)) \cong \mathrm{Hom}_{\mathcal{C}}(X, GF(X)).$$

2. for every $Y \in \mathcal{D}$, the morphism $\epsilon_Y : FG(Y) \to Y$ (each called a *counit morphism*) in $\mathcal{D}$ is obtained as the image of $\mathrm{id}_{G(Y)}$ via the adjoint isomorphism
$$\mathrm{Hom}_{\mathcal{C}}(G(Y), G(Y)) \cong \mathrm{Hom}_{\mathcal{D}}(FG(Y), Y).$$

Definition A.0.2. Let $\mathcal{C}$ and $\mathcal{D}$ be (large)) categories. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor.

1. F is called *full* if for every pair of objects $x, y \in \mathrm{Ob}(\mathcal{C})$, the induced rule/assignment/class function
$$F_{x,y} : \mathrm{Hom}_{\mathcal{C}}(x, y) \to \mathrm{Hom}_{\mathcal{D}}(F(x), F(y))$$
on Hom-collections is "surjective", i.e. for all morphisms $g : F(x) \to F(y)$, there exists some morphism $f : x \to y$ such that $F(f) = g$.
2. F is called *faithful* if for every pair of objects $x, y \in \mathrm{Ob}(\mathcal{C})$, the induced class function (assignment)
$$F_{x,y} : \mathrm{Hom}_{\mathcal{C}}(x, y) \to \mathrm{Hom}_{\mathcal{D}}(F(x), F(y))$$
on Hom-collections is "injective", i.e., for any morphisms $f_1, f_2 \in \mathrm{Hom}_{\mathcal{C}}(x, y)$, if $F(f_1) = F(f_2)$ in $\mathrm{Hom}_{\mathcal{D}}(F(x), F(y))$, then $f_1 = f_2$.
3. F is called *fully faithful* if it is both full and faithful.

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